

Exam 1 Correction

Thursday, October 1, 2026

10:16 PM

3. Consider the fixed point iteration

$$x_{n+1} = g(x_n)$$

where $x_0 \in [a, b]$ and $g(x)$ has a bounded first derivative (i.e. there exist a constant M such that $|g'(x)| \leq M \quad \forall x \in [a, b]$). Assume there exist a fixed point $p \in [a, b]$.

(a) Show that $|x_{n+1} - p| \leq M|x_n - p|$.

(b) Find a function $L(n)$ (it may also involve M) such that

$$|x_{n+1} - p| \leq L(n)|x_0 - p|.$$

(c) What conditions must be placed on the function $L(n)$ that you found in part (b) so that the fixed point iteration converges to p ?

a) I got Part a Correct...

b) $|x_{n+1} - p| \leq M|x_n - p|$
 $|x_n - p| \leq M|x_{n-1} - p|$

$\Rightarrow$

$$|x_{n+1} - p| \leq M^2|x_{n-1} - p|$$

following this pattern...

$$|x_{n+1} - p| \leq M^{n+1}|x_n - p|$$

$$\Rightarrow L(n) = M^{n+1}$$

$$c) \quad 0 \leq |x_{n+1} - p| \leq M |x_0 - p|$$

Squeeze theorem guarantees that fixed point iteration converges if...

$$\lim_{n \rightarrow \infty} L(n) |x_0 - p|$$

$$= |x_0 - p| \lim_{n \rightarrow \infty} L(n)$$

$$\Rightarrow \lim_{n \rightarrow \infty} L(n) = 0$$

↙ This is the requirement on $L(n)$
this occurs when $|M| < 1$

d) I also got part d correct.